

Helix OTA — WCAG-AA

OpenDesign Token Re-Vendor + Re-Prove

Revision: 1 **Last modified:** 2026-07-09T18:30:01Z **Scope:** design-systems/helix-ota/tokens.css (canonical) vendored byte-identical into clients/ota-manager/src/styles/opendesign-tokens.css + dashboard/src/styles/tokens.css. **Authority:** applies the accessibility fixes proposed (but not applied) in docs/research/opendesign_token_contrast_audit_20260709/CONTRAST.md. **Boundary (§11.4.6):** every ratio below is a real computed WCAG 2.1 number from the actual new hex (sRGB → linearize → luminance → ratio), and every render claim is backed by a captured host-render pass. No guessing.

1. Decision — final values (measured, not guessed)

Systematic-debugging finding (§11.4.102) that shaped the values: the dashboard renders --danger / --warn / --success as **small bold TEXT** — status badges (fontSize:12, fontWeight:600) and error/status messages (13px) in dashboard/src/components/ui.tsx + several screens — not only as color-mix(...) tint fills. Therefore the **4.5:1 normal-text bar (SC 1.4.3)** governs their text usage, which is stricter than the audit's provisional 3:1 “status/icon” classification. Because component source is out of scope (cannot wire a separate text-tone token), the **existing token values themselves** must be text-usable. The chosen tones are the Tailwind-family tones the dashboard originally used for this text before the vendoring flattened them (visible in the inline // #854d0e / // #166534 / // #991b1b comments), so the fix is on-brand by construction.

Two proposed values from the audit were escalated one Tailwind step because the proposed value **fails 4.5 on the real rendered background** (measured, below):

Finding	Token / theme	Old	Audit proposed	APPLIED	Why not the proposed value
M1a/ M1b	--danger DARK	#7f1d1d	#dc2626	#ef4444 (red-500)	#dc2626 as text = 4.14 on #020817 / 3.82 on its badge- tint — FAILS 4.5. #ef4444 = 5.32 / 4.72 PASS.
M1c	--warn LIGHT	#eab308	#a16207	#854d0e (amber-800)	#a16207 as text = 4.49 on warm / 4.03 on its badge- tint — FAILS 4.5. #854d0e = 6.85 / 5.46 PASS.
—	-- success LIGHT	#16a34a	(darker text tone)	#166534 (green-800)	green-700 #15803d badge- tint = 4.09 (<4.5); #166534 = 5.66 PASS.
F4	--muted LIGHT	#64748b	#475569	#475569 (slate-600)	proposed value passes (6.92 warm).
F5	-- border- strong	n/a	#64748b	#64748b (slate-500)	proposed value passes 3:1

Finding	Token / theme	Old	Audit proposed	APPLIED	Why not the proposed value
	(NEW, both themes)				all surfaces.

Structural note: `--warn` and `--success` were defined only in `:root` (light) and inherited by dark. Darkening them for light would have broken dark (dark amber/green on near-black), so **explicit dark overrides** were added pinning dark `--warn=#eab308` (10.43:1) and dark `--success=#16a34a` (6.07:1) — dark pixels for these are unchanged. `--muted` and `--danger` already had independent dark overrides.

2. Computed ratios — every changed token now meets AA (literal script output)

Script: `contrast_final.py` (committed beside this file). Thresholds: text 4.5:1 (SC 1.4.3), UI boundary 3.0:1 (SC 1.4.11). “`badgeTint`” = the token over its own `color-mix(..., transparent 85%) fill`; “`alertTint`” = over transparent 92%.

```

LIGHT --muted #64748b -> #475569 (text 4.5)
  #475569 thr4.5: bg=7.58✓ warm=6.92✓
LIGHT --warn #eab308 -> #854d0e (text 4.5)
  #854d0e thr4.5: bg=6.85✓ warm=6.25✓ badgeTint=5.46✓
  alertTint=6.09✓
LIGHT --success #16a34a -> #166534 (text 4.5)
  #166534 thr4.5: bg=7.13✓ warm=6.51✓ badgeTint=5.66✓
DARK --danger #7f1d1d -> #ef4444 (text 4.5; dark surface #020817)
  #ef4444 thr4.5: surface=5.32✓ badgeTint=4.72✓ alertTint=5.05✓
NEW --border-strong #64748b (UI boundary 3.0; both themes)
  #64748b thr3.0: L/bg=4.76✓ L/warm=4.34✓ D/bg=4.20✓ D/
  warm=3.07✓
DARK-PRESERVATION overrides (must still PASS)
  dark --warn #eab308: surface=10.43✓ badgeTint=8.31✓
  dark --success #16a34a: surface=6.07✓ badgeTint=5.21✓

```

Every changed token is now $\geq$ **its threshold** on every surface it renders on — under BOTH the audit’s 3:1 UI bar AND the stricter 4.5:1 text bar (the applied values clear 4.5).

Honest residual (§11.4.6)

--danger LIGHT (#dc2626, **unchanged** — **out of scope** this round) is 4.83:1 on white (passes 4.5 plain text, passes the audit's 3:1) but 3.81:1 on its own 15% badge-tint. It was not flagged by the audit and is not in this round's change set; recorded here honestly as a follow-up rather than silently expanded into (§11.4.6, "do not over-change").

3. Byte-identity of the three vendored copies (cmp proof)

```
$ cmp design-systems/helix-ota/tokens.css clients/ota-manager/src/
styles/opendesign-tokens.css -> IDENTICAL
$ cmp design-systems/helix-ota/tokens.css dashboard/src/styles/
tokens.css -> IDENTICAL
$ cmp clients/ota-manager/src/styles/opendesign-tokens.css
dashboard/src/styles/tokens.css -> IDENTICAL
sha256 (all three) =
14a006da257643793e291536adac6e0649965e8369db766afb8a41c8e4fdcb86
```

(dashboard's copy is named `tokens.css`, ota-manager's `opendesign-tokens.css` — same bytes.)

4. Host-render re-proof (goldens regenerated, analyzers still self-validate)

ota-manager (pnpm hostrender — regen baseline + self-diff + OCR + layout, both themes): OVERALL: PASS. Each analyzer self-validated (golden-good passes AND golden-bad flagged): image-diff good=0.0000% / bad flagged (0.78% light, 1.52% dark); OCR all labels present / suppressed label flagged; layout clean / collapsed-submit flagged. `image_diff_analyzer_sound`, `layout_analyzer_sound`, `ocr_analyzer_sound` = **all true**.

- Determinism check (§11.4.6): a NEW-token render vs an OLD-token render of ota-manager Login = **0.0000%** both themes → the token change is **inert for ota-manager** (its `--muted/--border` collide with `shadcn` HSL tokens which win the cascade, and it does not consume `--danger/--warn/--success`). The 0.77% delta between the previously-committed baseline and the fresh render is pre-existing cross-build AA variance, not this change.

dashboard (npm run e2e:hostrender): - --update-snapshots regen pass: **45 passed**. - clean verify pass (no update): **45 passed** (audit/releases/artifact-upload/appshell × light+dark golden-good + layout-self-validated + image-diff-self-validated + baseline-rejects-mutated + light ↔ dark distinctness). The committed golden PNGs stayed within the config's maxDiffPixelRatio: 0.01 AA tolerance (the muted-text delta is sub-1% of pixels), so no golden bytes needed rewriting; the -actual.png evidence reflects the new render.

Screenshots of affected screens (still render correctly — no overlap/clip/collapse, legible text) under screenshots/: dashboard-releases-{light,dark}.png, dashboard-audit-{light,dark}.png, dashboard-artifact-upload-light.png, dashboard-login-light.png, ota-manager-login-{light,dark}.png.

5. Regression — builds + unit suites (all green)

Command	Result
dashboard: npm run build (tsc×3 + vite)	exit 0
dashboard: npm run test:run (vitest)	107 passed (12 files)
ota-manager: pnpm build (vite)	exit 0
ota-manager: vitest run	36 passed (9 files)

ota-manager's 118 pre-existing tsc type errors (separate tracked item) are unrelated to this change and were not touched; vite build + vitest are unaffected.

6. Files changed

- design-systems/helix-ota/tokens.css (canonical) + 2 byte-identical vendored copies.
- Regenerated host-render evidence under docs/qa/20260709-ota-manager-hostrender/ and docs/qa/20260709-dashboard-* (-actual.png refresh; committed goldens unchanged, within tolerance).
- This evidence dir + docs/research/opendesign_token_contrast_audit_20260709/CONTRAST.md (marked APPLIED, revision bumped).

Sources verified

Sources verified 2026-07-09: - WCAG 2.1 SC 1.4.3 Contrast (Minimum) — <https://www.w3.org/WAI/WCAG21/Understanding/contrast-minimum.html> - WCAG 2.1 SC 1.4.11 Non-text Contrast — <https://www.w3.org/WAI/WCAG21/Understanding/non-text-contrast.html> - WCAG relative-luminance + contrast-ratio — <https://www.w3.org/TR/WCAG21/#dfn-relative-luminance> / [#dfn-contrast-ratio](https://www.w3.org/TR/WCAG21/#dfn-contrast-ratio)